VULTR AGENT RUSH · BLAST RADIUS ZERO TRACK

Blast Radius Zero

An agent whose every action runs inside a throwaway Docker sandbox on a Vultr VM.

It plans in JSON, writes Python, runs it in a fresh container, and returns the real stdout.

Containment is measured, not asserted: rm -rf / is blocked, egress fails, the host survives.

github.com/Hantsuo/blast-radius-zero

WHY THIS EXISTS

The problem

Every 'agent that does real work' ends up running code nobody wrote: model output, scraped snippets, user CSVs.

One bad import and it deletes a production disk, exfiltrates a key, or forks until the host stops answering.

Chat-only agents are demos. Agents that execute need a blast radius of zero.

FOUR STEPS

How it works

1 Plan — the model receives the task and returns JSON: a thought plus Python.

2 Dispatch — the supervisor starts a fresh container per step; the app process never imports user code.

3 Contain — --network none, read-only root, 512 MB, 1 CPU, 128 PIDs, cap-drop ALL, 15 s wall clock.

4 Verify — stdout, exit code and duration return to the UI; the container is destroyed.

EVIDENCE FROM THE SUITE

Measured containment

rm -rf / + os.remove('/etc/passwd') → BLOCKED (Read-only file system), host sentinel intact, 840 ms

while True: pass → killed at the wall clock, 3.2 s, timed_out=true

socket.create_connection('1.1.1.1', 53) → NETWORK BLOCKED, 392 ms

host environment probe → CLEAN; fork bomb → BlockingIOError at the PID limit

docker ps afterwards → empty

WHAT RUNS WHERE

Vultr integration

Vultr Cloud Compute: FastAPI orchestrator, demo UI and sandbox supervisor (Ubuntu 24.04, 2 vCPU / 4 GB, Tokyo).

Vultr Serverless Inference: the planner, OpenAI-compatible at api.vultrinference.com/v1.

Sandbox: sibling Docker containers on the same VM, created and destroyed per task.

No GPU: inference is remote, execution is CPU-only.

WHO PAYS FOR THIS

Business value

Support and ops teams running model-written scripts against real data.

Platform teams that need a code-execution primitive with a measurable blast radius.

The red-team suite doubles as a CI gate: any new escape attempt becomes a test.

Live demo: http://167.179.96.32:8000/healthz